

Xiyuan (TOMMY) GUO

tommyguo024@outlook.com — (516) 474-3279 — xguo.info — github.com/xguot — linkedin.com/in/xiyuan-tommy-guo

EDUCATION

University of Virginia

Charlottesville, VA — Sept 2024 - May 2028 (Expected)

Degree: Bachelor of Arts in Computer Science and Mathematics — GPA 3.89/4.00

Relevant Courses: Computer Systems and Organization 2, Discrete Math and Theory, Data Structures and Algorithms, etc.

PROFESSIONAL EXPERIENCE

Full-Stack Developer, UVA (Ke's Group) (Next.js, SQL, PHP)

Charlottesville, VA — Oct 2025 - Present

- Developed a comprehensive full-stack research platform utilizing Next.js framework and PHP backend, architecting a unified SQL database to catalog rare earth element properties and facilitate global academic data retrieval.
- Engineered a custom data integration pipeline using Go scripts to aggregate high-dimensional material parameters and JSON machine learning outputs into a centralized relational database, streamlining the transition from raw research to production.
- Created immersive data visualizations using the Three.js rendering engine with JSmol viewer generating interactive 3D atomic models with Kiviat diagram to enable real-time structural inspection and analysis by investigators globally.
- Orchestrated high-availability Linux web infrastructure and server-side security, integrating OAuth 2.0 and custom Access Control (RBAC) layers to enforce multi-tenant permissions and guarantee data integrity across the platform's global portal.

Research Assistant, Yale University (Liu's Group) (ML, CNN, PyTorch)

New Haven, CT — Feb 2026 – Present

- Developing specialized U-Net architectures in PyTorch, integrating physics-informed constraints to reconstruct high-fidelity 3D volumes from raw computational sensor data and depth-dependent point spread functions (PSFs).
- Engineered high-concurrency 3D data voxelization and physics-based simulation engines in Go, achieving a 50x performance increase in preprocessing throughput to significantly accelerate the end-to-end training cycle for large-scale deep learning models.
- Solving complex inverse problems in computational imaging by implementing high-dimensional optimization frameworks, specifically leveraging ADMM and Augmented Lagrange Multipliers to optimize reconstruction accuracy.
- Evaluating the architectural efficacy of novel Convolutional Physics-Informed Neural Networks by benchmarking performance against diverse datasets, ensuring model robustness and reconstruction quality compared to existing computational baselines.
- Engaging in collaboration with researchers at Yonsei University to incorporate specialized Point Spread Function datasets, facilitating advanced optical modeling to optimize model convergence for apertures and sensor configurations.

Quantitative Analyst Intern, Hangzhou Landscaping

Hangzhou, China — Jun 2024 - Aug 2024

- Engineered quantitative financial models using Python (Pandas, NumPy) to evaluate over \$500,000 in procurement assets, automating cash-flow simulations and ROI analysis to optimize project viability and transparency.
- Developed a comprehensive risk-management framework by identifying, assessing, and monitoring key operational risks across 10,000+ transaction records; incorporated statistical scoring models and scenario-based sensitivity analysis.
- Architected a high-throughput financial pipeline in Go, modeling asset price movements via normal distributions to compute VaR and optimize capital allocation. Engineered concurrent simulation algorithms to drastically reduce computational latency, validating the models through rigorous statistical backtesting to ensure resilient returns under high market volatility.

PROJECTS

Creator & Lead Maintainer, difi (Go, Bubble Tea)

Dec 2025 - Present

- Engineered a Go-based Git diff TUI using Bubble Tea and Neovim, optimizing developer productivity by streamlining complex version control and facilitating intuitive git diff for code reviews before Pull Requests.
- Spearheaded an active open-source ecosystem through 90+ commits to release v1.0.10, managing issue triage and integrating global pull requests; drove adoption to 320+ GitHub stars and secured #6 on Hacker News.
- Achieved official inclusion into the Homebrew core repository, navigating stringent maintainer reviews to streamline cross-platform distribution for macOS and Linux, effectively scaling adoption to over 1,000 active developers.

Open-Source Contributor, Neovim (Lua, C)

May 2025 - Present

- Engineered high-performance components Lua-based Neovim optimizations, navigating strict code style guidelines and rigorous PR review processes to contribute low-latency workflow enhancements to an ecosystem of over 100,000 active developers.

Creator & Lead Engineer, zswEEP (TypeScript, SvelteKit, Git)

Dec 2025 - Present

- Engineered a zero-latency Vim-emulated Minesweeper engine with a custom FSM and Supabase; integrated global community assets across 200+ commits, scaling to 10,000+ users and 700+ gameplay hours during a #5 Hacker News launch.

CERTIFICATIONS & SKILLS

- **Languages:** Go, Python, C/C++, TypeScript/JavaScript, Java, Lua, Bash, R, Ruby, Lean, SQL, LaTeX
- **Frameworks & Libraries:** PyTorch, TensorFlow, React, Next.js, SvelteKit, Bubble Tea, Pandas
- **Developer Tools & Cloud:** AWS (Beanstalk, Aurora), CUDA, QEMU, HPC/Slurm, Supabase, Make, Git, Nvim/tmux
- **Certifications:** AWS Certified AI Practitioner, AWS Certified Cloud Practitioner, USACO Silver

LEADERSHIP

Firefighter, Plainfield Fire Department

Plainfield, NH — Sept 2022 - Apr 2024

- Executed rapid situational assessments and critical decision-making across 100+ annual high-pressure emergency responses.